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MATH3311/MATH5335

Topic 09B – The binomial option pricing model (BOPM)

BOPM: is an options valuation method

- BOPM: invented by Cox-Ross-Rubinstein in 1979. It was originally invented as a tool to explain the Black-Scholes Model.
- BOPM is a widely used numerical simulation for valuing options. It provides a discrete-time framework for evaluating the price of an option over multiple time steps. The BOPM is more accurate for American Style Options.
- The BOPM assumes that stock price have two possible movement directions at each time level: up or down
- The BOPM produces a binomial distribution (Binomial Tree) of all the possible paths that a stock price could take during the life of the option.
- The BOPM assumes that stock price have two possible movement directions at each time level: up or down

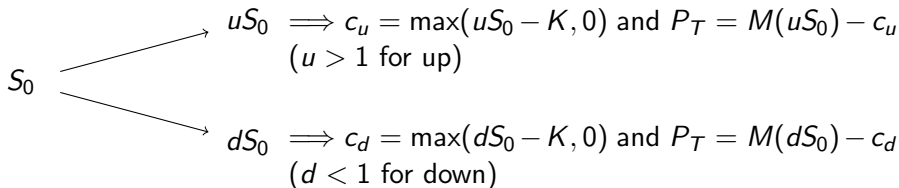
Valuation of a call option using BOPM

- Recall that, a **European call option** on an underlying asset gives the owner the right (but not the obligation) to **buy** that asset for a nominated **strike price** K at a nominated **expiry** time T .
- Given that: S_0 is the price of underlying asset (at the current time) and r is the interest rate
- Consider a portfolio that consisted of long position in M stocks and short position in **one** written (selling) call option contract. If c represents the value of owning the call then the liability involve with having written the call is $-c$. So, the current value of the portfolio is $P_0 = MS_0 - c$.
- For a complete BOPM, we require **A1** and **A2** in the next slide.

One time-step binomial tree

$$t_0 = 0$$

$$t_1 = T$$



- **A1:** Choose M so that the value of the portfolio at the expiry time, P_T , is the same in each case. That is,

$$M(uS_0) - c_u = M(dS_0) - c_d \implies M = \frac{c_u - c_d}{S_0(u - d)}$$

- **A2:** Assume that P_0 equals time discounted value of the P_T using risk-free interest rate r (prices do not allow arbitrage). So,

$$P_0 = e^{-rT} P_T \implies$$

$$MS_0 - c = e^{-rT} (M(uS_0) - c_u) \implies c = MS_0(1 - e^{-rT}u) + e^{-rT}c_u$$

One time-step binomial tree

$$M = \frac{c_u - c_d}{S_0(u - d)} \quad \text{and} \quad c = MS_0(1 - e^{-rT}u) + e^{-rT}c_u$$

$$\begin{aligned} c &= \frac{c_u - c_d}{u - d}(1 - e^{-rT}u) + e^{-rT}c_u \\ &= \frac{e^{-rT}}{u - d} \left((c_u - c_d)(e^{rT} - u) + c_u(u - d) \right) \\ &= \frac{e^{-rT}}{u - d} \left(c_u(e^{rT} - d) - c_d(e^{rT} - u) \right) \\ &= e^{-rT} \left(c_u \frac{e^{rT} - d}{u - d} + c_d \frac{u - e^{rT}}{u - d} \right) \end{aligned}$$

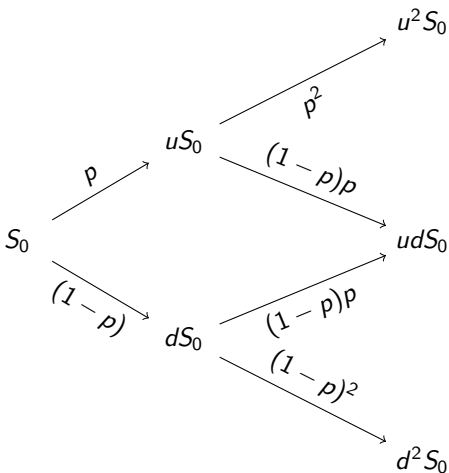
Assume that $d \leq e^{rT} \leq u$ with $d < u$, and then

$$c = e^{-rT} \left(c_u p + c_d(1 - p) \right), \quad \text{with} \quad p = \frac{e^{rT} - d}{u - d}$$

Two time-step binomial tree

$$t_0 = 0 \quad t_1 = \delta t$$

$$t_2 = 2\delta t$$



Assume that $d \leq e^{r\delta t} \leq u$

$p = \frac{e^{r\delta t} - d}{u - d}$. For $j = 0, 1, 2$,

$$c_{u^{2-j}d^j} = \max\{u^{2-j}d^j S_0 - K, 0\}$$

$$c = e^{-r\delta t} (p c_u + (1-p) c_d)$$

$$c_u = e^{-r\delta t} (p c_{u^2} + (1-p) c_{ud})$$

$$c_d = e^{-r\delta t} (p c_{du} + (1-p) c_{d^2})$$

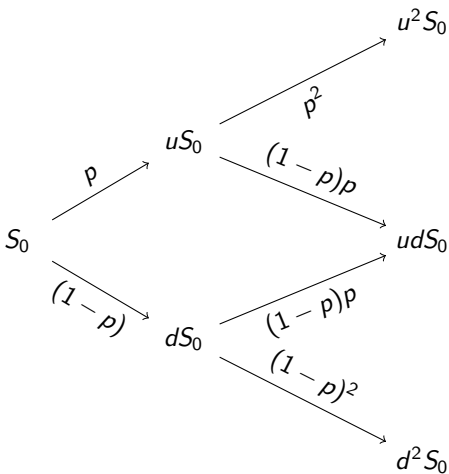
$$c = e^{-2r\delta t} \times (p^2 c_{u^2} + 2p(1-p) c_{ud} + (1-p)^2 c_{d^2})$$

$$c = e^{-rT} \sum_{j=0}^2 \binom{2}{j} p^{2-j} (1-p)^j c_{u^{2-j}d^j}$$

Two time-step binomial tree- Alternative representations

$$t_0 = 0 \quad t_1 = \delta t$$

$$t_2 = 2\delta t$$



Let $C^{i,j} = c_{u^i d^j}$. For $j = 0, 1, 2$,
 $C^{2-j,j} = \max\{u^{2-j} d^j S_0 - K, 0\}$

Tree: $C^{0,0}$ $C^{1,0}$ $C^{2,0}$
 $C^{0,1}$ $C^{1,1}$
 $C^{0,2}$

$$c_u = C^{1,0} = e^{-r\delta} \left(p C^{2,0} + (1-p) C^{1,1} \right)$$

$$c_d = C^{0,1} = e^{-r\delta} \left(p C^{1,1} + (1-p) C^{0,2} \right)$$

$$C^{0,0} = e^{-2r\delta} \left(p^2 C^{2,0} + 2p(1-p) C^{1,1} + (1-p)^2 C^{0,2} \right)$$

$$c = C^{0,0} = e^{-rT} \sum_{j=0}^2 \binom{2}{j} p^{2-j} (1-p)^j C^{2-j,j}$$

General formula for N time-step binomial tree

Set $\delta t = T/N$ and $t_i = i\delta t$ for $0 \leq i \leq N$. Then, at time node t_i ,

$$C^{i-j,j} = e^{-r\delta t} \left(p C^{i-j+1,j} + (1-p) C^{i-j,j+1} \right) \quad \text{for } 0 \leq j \leq i < N,$$

with $C^{N-j,j} = \max\{u^{N-j}d^j S_0 - K, 0\}$, $j = 0, 1, \dots, N$. In a series form,

$$c = C^{0,0} = e^{-rT} \sum_{j=0}^N \binom{N}{j} p^{N-j} (1-p)^j C^{N-j,j}$$

where

$$\binom{N}{j} = \frac{N!}{j!(N-j)!} \quad \text{and} \quad p = \frac{e^{r\delta t} - d}{u - d}$$

With σ being the volatility, Cox-Ross-Rubinstein (1979) suggest

$$u = e^{\sigma\sqrt{\delta t}} \quad \text{and} \quad d = e^{-\sigma\sqrt{\delta t}}$$

N time-step binomial tree - Implementations

Set $\delta t = T/N$ and $t_i = i\delta t$ for $0 \leq i \leq N$. Then, at time node t_i ,

$$C^{i-j,j} = e^{-r\delta t} \left(p C^{i-j+1,j} + (1-p) C^{i-j,j+1} \right) \quad \text{for } 0 \leq j \leq i < N, \quad (1)$$

with $C^{N-j,j} = \max\{u^{N-j} d^j S_0 - K, 0\}$, $j = 0, 1, \dots, N$.

Practical implementations: for $i = 0, 1, \dots, N$, the $(i+1)$ -by-1 column vector

$$\mathbf{C}^i = [C^{i,0}, C^{i-1,1}, \dots, C^{0,i}]^T.$$

Extract the i -by-1 column vectors $\mathbf{C}_a^i$ and $\mathbf{C}_b^i$ from $\mathbf{C}^i$ by dropping out the last component and first component of $\mathbf{C}^i$, respectively.

- (1) is equivalent to

$$\mathbf{C}^i = e^{-r\delta t} \left(p \mathbf{C}_a^{i+1} + (1-p) \mathbf{C}_b^{i+1} \right) \quad \text{for } 0 \leq i < N$$

- $c = C^{0,0} = \mathbf{C}^0$.

Example:

Compute the value of a European call option using a binomial tree model with $N = 300$ time steps. Given that $S_0 = 100$, $r = 0.1$, $\sigma = 0.2$, $K = 107$ and $T = 2$ (years). So, $\delta t = T/N = 1/150$. **Answer:** $c = 17.8711$

- Matlab:

```
j = 0:N;
C(:,N+1) = max(0,u.^(N-j).*d.^j.*S0-K); %at T
for i = N:-1:1
    C(1:i,i) = exp(-r*dt)*(p*C(1:i,i+1) + (1-p)*C(2:(i+1),i+1));
end
c = C(1,1)
```

- Python:

```
C = zeros((N+1,N+1))
j = arange(0,N+1)
C[:,N] = maximum(0,u**(N-j)*d**j*S0-K)
for i in arange(N,0,-1):
    C[0:i,i-1] = exp(-r*dt)*(p*C[0:i,i] + (1-p)*C[1:(i+1),i])
c = C[0,0]
```

BOPM for valuing American (call) options

- An **American call option** on an underlying asset gives the owner the right (but not the obligation) to **buy** that asset for a nominated **strike price** K at any time before a nominated **expiry** time T .
- Set $\delta t = T/N$ and $t_i = i\delta t$ for $0 \leq i \leq N$.
- At each time node t_i , the value of American call option is given by

$$\tilde{C}^{i-j,j} = \max \left\{ e^{-r\delta t} (p\tilde{C}^{i-j+1,j} + (1-p)\tilde{C}^{i-j,j+1}), \max\{u^{i-j}d^j S_0 - K, 0\} \right\},$$

for $0 \leq j \leq i < N$.

- $$\tilde{C}^{N-j,j} = \max\{u^{N-j}d^j S_0 - K, 0\} \text{ for } j = 0, 1, \dots, N.$$
- In a vector form, with $\mathbf{B}^i = [\max\{u^{i-j}d^j S_0 - K, 0\}]^T$, (($i+1$)-by-1 vector)

$$\tilde{\mathbf{C}}^i = \max \left(e^{-r\delta t} (p\tilde{\mathbf{C}}_a^{i+1} + (1-p)\tilde{\mathbf{C}}_b^{i+1}), \mathbf{B}^i \right) \text{ for } 0 \leq i < N$$

- $c = \tilde{C}^{0,0} = \tilde{\mathbf{C}}^0$ (estimated value of the American call option)
- Based on the given data in the last example, $c = 25.1592$